

SMART DAIRY

AI-Powered Dairy Management Platform

Software Name:	Smart Dairy Platform
Version:	1.0.0
Release Date:	October 22, 2025
Authors:	Smart Dairy Development Team
License:	Proprietary Software
Platform:	Web Application (Django)
AI Engine:	Google Gemini 2.0 Flash
Database:	SQLite/PostgreSQL

ABSTRACT

Purpose:

Smart Dairy revolutionizes dairy farming through AI-powered health monitoring, disease detection, feed optimization, and production analytics.

Core Technologies:

• Python Django 5.2.7 Framework • Google Gemini 2.0 Flash AI Engine • Computer Vision for Disease Detection • Natural Language Processing (English/Swahili) • Bootstrap 5.3.0 Responsive UI • SQLite/PostgreSQL Database • RESTful API Architecture

Deployment Platform:

Cloud-native web application supporting AWS, Google Cloud, Azure deployment with mobile-responsive interface.

SYSTEM ARCHITECTURE

3.1 Database Models

```
# Complete Cow Management Model
class Cow(models.Model):
    HEALTH_CHOICES = [
        ('excellent', 'Excellent'),
        ('good', 'Good'),
        ('fair', 'Fair'),
        ('poor', 'Poor'),
        ('sick', 'Sick')
    ]
    owner = models.ForeignKey(User, on_delete=models.CASCADE)
    name = models.CharField(max_length=100)
    tag_number = models.CharField(max_length=50, unique=True)
    breed = models.CharField(max_length=100)
    age = models.IntegerField()
    weight = models.FloatField()
    health_condition = models.CharField(max_length=20, choices=HEALTH_CHOICES)
    date_added = models.DateTimeField(default=timezone.now)

    class DailyFeedRecord(models.Model):
        cow = models.ForeignKey(Cow, on_delete=models.CASCADE)
        date = models.DateField()
        protein_feed = models.FloatField()
        silage = models.FloatField()
        minerals = models.FloatField()
        milk_yield = models.FloatField(null=True, blank=True)
        is_ai_recommended = models.BooleanField(default=False)
        week_start_date = models.DateField(default=timezone.now)
```

3.2 AI Service Engine

```
class AIService:
    def __init__(self):
        genai.configure(api_key=settings.AI_API_KEY)
        self.model = genai.GenerativeModel('gemini-2.0-flash-exp')
        self.vision_model = genai.GenerativeModel('gemini-2.0-flash-exp')

    def analyze_disease_photo(self, image_file, language='en'):
        image = Image.open(image_file)
        prompt = self._build_disease_analysis_prompt(language)
        response = self.vision_model.generate_content([prompt, image])
        return self._parse_disease_response(response.text, language)

    def get_feed_recommendation(self, cow_data, language='en'):
        prompt = self._build_feed_prompt(cow_data, language)
        response = self.model.generate_content(prompt)
        return self._parse_feed_response(response.text, language, cow_data)

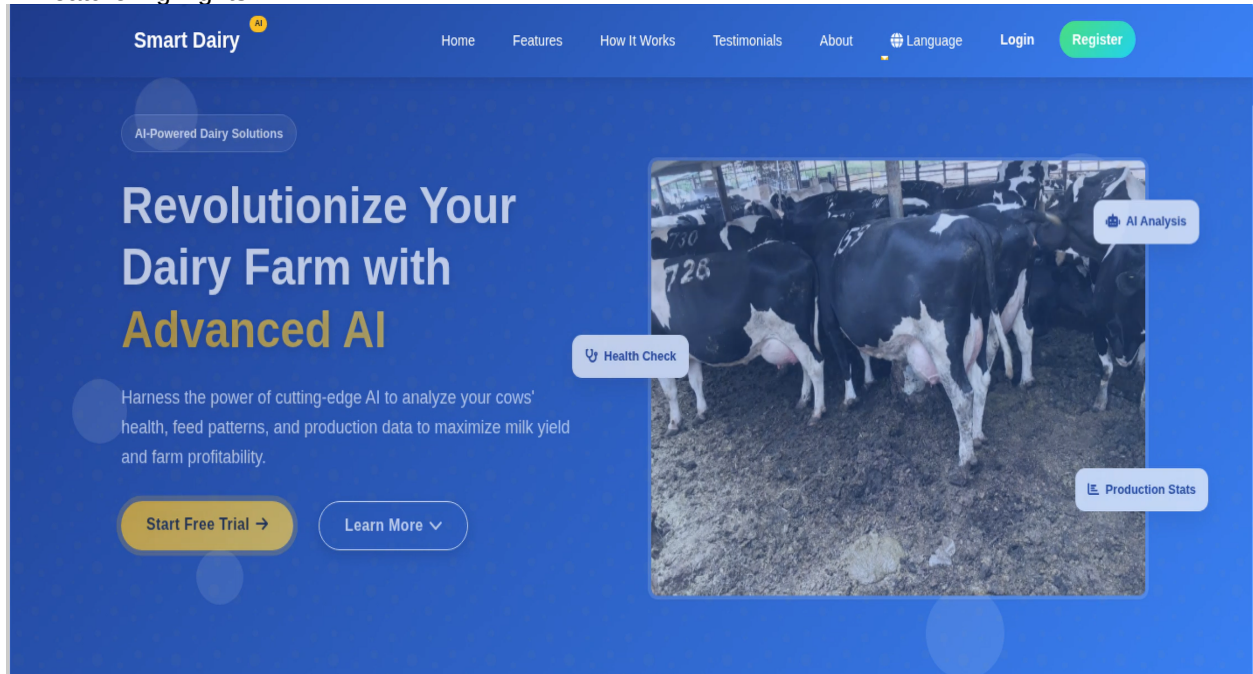
    def chat_response(self, message, language='en'):
        simple_prompt = f"Answer this farming question: {message}"
        response = self.model.generate_content(simple_prompt)
        return response.text.strip() if response.text else "No response"

    def predict_milk_production(self, cow_data, language='en'):
        prompt = self._build_prediction_prompt(cow_data, language)
        response = self.model.generate_content(prompt)
        return self._parse_prediction_response(response.text, language)
```


4. USER INTERFACE DOCUMENTATION

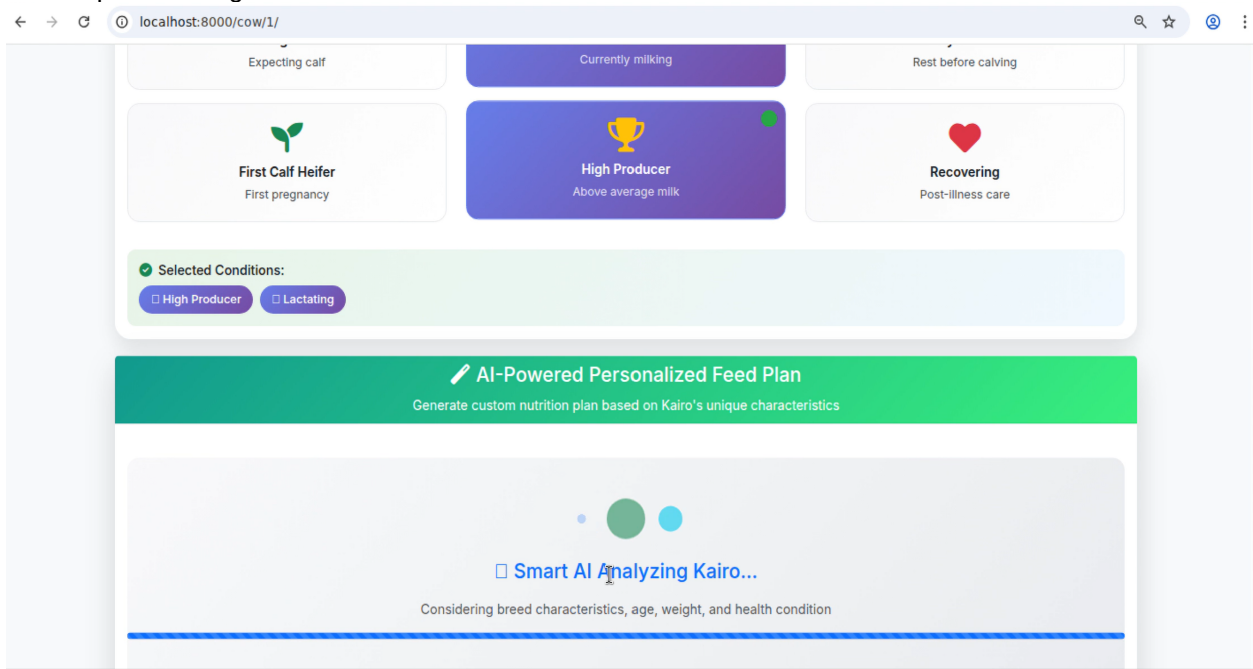
4.1 Main Landing Interface

The main interface showcases AI-powered dairy solutions with comprehensive navigation and feature highlights.



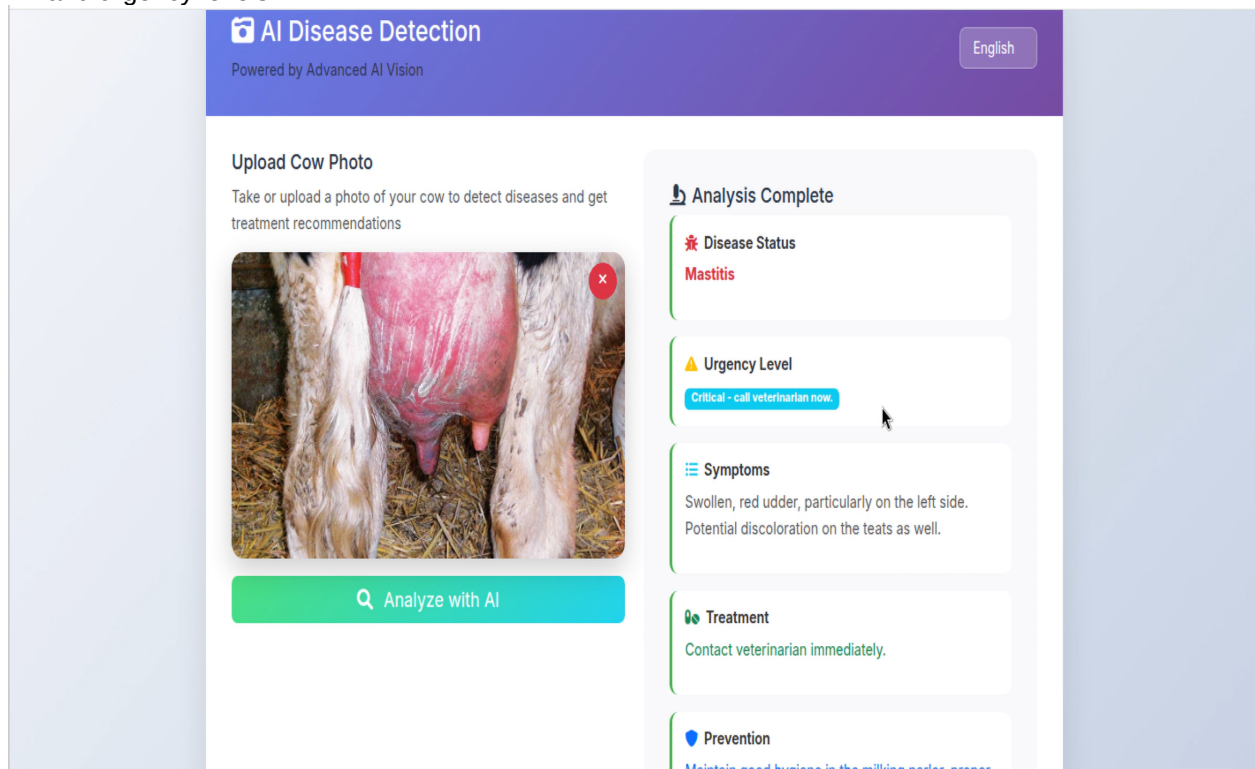
4.2 AI Feed Personalization System

Advanced AI generates personalized nutrition plans based on cow characteristics, health status, and production goals.



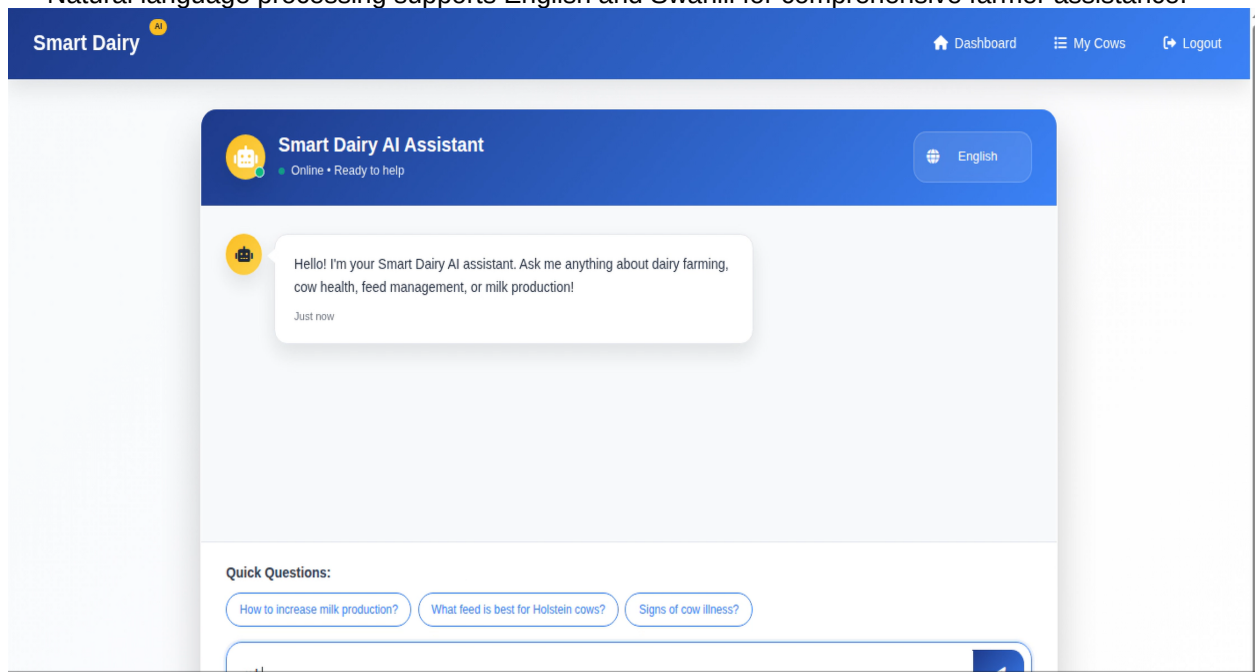
4.3 Computer Vision Disease Detection

AI-powered image analysis provides instant disease diagnosis with treatment recommendations and urgency levels.



4.4 Bilingual AI Assistant

Natural language processing supports English and Swahili for comprehensive farmer assistance.



5. TECHNICAL IMPLEMENTATION

5.1 Core Application Views

```
@login_required def dashboard(request): user_cows = Cow.objects.filter(owner=request.user) total_cows = user_cows.count() context = { 'total_cows': total_cows, 'cows': user_cows[:5], 'recent_records': DailyFeedRecord.objects.filter( cow__owner=request.user ).order_by('-date')[:10] } return render(request, 'main/dashboard.html', context) def ai_disease_detection(request): if request.method == 'POST' and request.FILES.get('cow_image'): image = request.FILES['cow_image'] language = request.POST.get('language', 'en') ai_service = AIService() analysis = ai_service.analyze_disease_photo(image, language) return JsonResponse(analysis) return render(request, 'main/disease_detection.html') def generate_feed_plan(request, cow_id): cow = get_object_or_404(Cow, id=cow_id, owner=request.user) ai_service = AIService() cow_data = { 'breed': cow.breed, 'weight': cow.weight, 'age': cow.age, 'health_condition': cow.health_condition } feed_plan = ai_service.get_feed_recommendation(cow_data) return JsonResponse(feed_plan)
```

5.2 URL Routing Configuration

```
urlpatterns = [ path('', views.landing_page, name='landing'), path('register/', views.register_view,
name='register'), path('login/', views.login_view, name='login'), path('dashboard/', views.dashboard,
name='dashboard'), path('add-cow/', views.add_cow, name='add_cow'), path('cow/', views.cow_detail,
name='cow_detail'), path('ai-analysis/', views.ai_analysis, name='ai_analysis'), path('feed-plan/',
views.generate_feed_plan, name='feed_plan'), path('disease-detection/', views.disease_detection,
name='disease_detection'), path('chat/', views.ai_chat, name='ai_chat'), path('production-stats/',
views.production_stats, name='production_stats'), ]
```

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